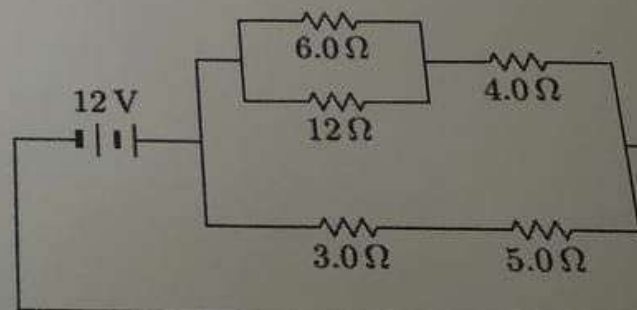


Part I: Choose the correct answer from the given alternatives and write your answer in the space provided (7 %).

1. Let $\vec{A} = (2m)\hat{i} + (6m)\hat{j} - (3m)\hat{k}$ and $\vec{B} = (4m)\hat{i} + (2m)\hat{j} + (1m)\hat{k}$. Then $\vec{A} \cdot \vec{B} =$
 - A. $(8m)\hat{i} + (12m)\hat{j} - (3m)\hat{k}$
 - B. $(12m)\hat{i} - (14m)\hat{j} - (20m)\hat{k}$
 - C. $23 m^2$
 - D. $17 m^2$
2. The inertia of a body tends to cause the body to:
 - A. speed up
 - B. slow down
 - C. resist any change in its motion
 - D. fall toward Earth
3. An elastic collision is one in which:
 - A. momentum is not conserved but kinetic energy is conserved
 - B. total mass is not conserved but momentum is conserved
 - C. kinetic energy and momentum are both conserved
 - D. momentum is conserved but kinetic energy is not conserved
4. A rock, which weighs 1400 N in air, has an apparent weight of 900 N when submerged in fresh water (998 kg/m^3). The volume of the rock is:
 - A. 0.14 m^3
 - B. 0.60 m^3
 - C. 0.90 m^3
 - D. $5.1 \times 10^{-2} \text{ m}^3$
5. A system undergoes an adiabatic process in which its internal energy increases by 20 J. Which of the following statements is true?
 - A. 20 J of work was done on the system
 - B. 20 J of work was done by the system
 - C. the system received 20 J of energy as heat
 - D. the system lost 20 J of energy as heat
6. A particle with charge $2\text{-}\mu\text{C}$ is placed at the origin, an identical particle, with the same charge, is placed 2 m from the origin on the x axis, and a third identical particle, with the same charge, is placed 2 m from the origin on the y axis. The magnitude of the force on the particle at the origin is:
 - A. $9.0 \times 10^{-3} \text{ N}$
 - B. $6.4 \times 10^{-3} \text{ N}$
 - C. $1.3 \times 10^{-2} \text{ N}$
 - D. $1.8 \times 10^{-2} \text{ N}$
7. The current in the $5.0\text{-}\Omega$ resistor in the circuit shown is:
 - A. 0.42 A
 - B. 0.67 A
 - C. 1.5 A
 - D. 2.4 A



Part III. Give a short and brief answer for the following questions. (8 %)

1. State:

a) the Zeroth law of thermodynamics

b) the first law of thermodynamics

2. State:

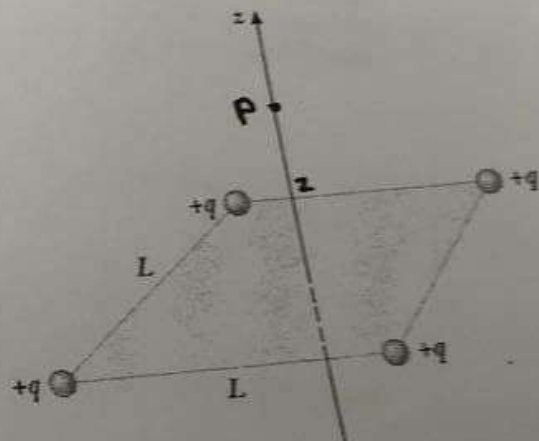
a) Faradays law of electromagnetic induction

b) Lenz's law

...ns clearly & Neatly.[25%]

4. Four identical charges, each having a charge of $+q = 4 \times 10^{-6} \text{ C}$, are fixed at the corners of a square of side $L = \sqrt{8} \text{ m}$.
- A) Find the electric field at a point P that lies at a distance $Z = 2 \text{ m}$ along the line perpendicular to the plane of the square and passing through the center of the square. (Fig. below) (3pts.)
 - B) Find the net electric force on a fifth point charge $+q = 4 \times 10^{-6} \text{ C}$, at a point P that lies at $Z = 2 \text{ m}$ above the center of the square due to the other four charges. (2pts.)
 - C) Calculate the potential energy due to the four charge's distribution located on the square (2pts.)

(use: $K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$)



Part II. Fill in the Blank Spaces with the most simplified Answers. [10 pts.]

1. If an ideal gas absorbs $5 \times 10^3 \text{ J}$ of energy while doing $2 \times 10^3 \text{ J}$ of work on the environment during a constant pressure process, then the change in the internal energy of the gas will be []
2. If identical resistors of 8Ω are connected in parallel combination, their equivalent resistance is 2Ω . The number of resistors in the circuit are []
3. At one instant an electron ($q = -1.6 \times 10^{-19} \text{ C}$) is moving in the XY-plane and the components of its velocity is $V_x = 5 \times 10^5 \text{ m/s}$ & $V_y = 3 \times 10^5 \text{ m/s}$. If magnetic field of 0.8 T is in +X-direction, then the magnitude of magnetic force is [] N.
4. If the temperature of a cube made of copper with initial volume 3 m^3 , changes from 10°C to 15°C , its change in volume is [] m^3 (use: $\alpha = 17 \times 10^{-6} \text{ K}^{-1}$)
5. A hydraulic press has one piston of diameter 2 cm & the other piston of diameter 8 cm . What force must be applied to the smaller piston to obtain a force of 1600 N at the larger piston? [] N

Part II: Fill the black spaces with the correct answer. [10pts]

1. In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 2 cm. This pressure is transmitted by a liquid to a piston that has a radius of 4 cm and 1.2×10^4 N is used to lift a car weight. The force that must be exerted is N **30000 N**
2. If the temperature of a cube made of copper with initial volume 8 m^3 changes from 10°C to 20°C , its change in volume is m^3 (Use: $\alpha = 1.7 \times 10^{-5} \text{ K}^{-1}$) **$4.08 \times 10^{-3} \text{ m}^3$**
3. A 1mol sample of an ideal gas is kept at 0°C during an expansion from 3 L to 10 L. The energy transferred by heat between the gas and the surroundings in this process is J **-2.7×10^3 J**
4. Two resistors connected in series have an equivalent resistance of 690Ω . When they are connected in parallel, their equivalent resistance is 150Ω . The resistance of two resistors are and **470Ω and 220Ω**
5. A conducting sphere has charge Q and its electric potential is V , relative to the potential far away. If the charge is doubled to $2Q$, the potential is **$2V$ or (doubled)**
 $V_2 = 2V_1$

Part III: Give a brief answer for the following questions. (8pts)

1. Define
 - a) latent heat of fusion
 - b) Latent heat of vaporization
2. State the following:
 - a) Faraday's law of induction
 - b) Lenz's law
3. List two heat transfer mechanisms and explain their difference